

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime behavior, and deployment.

1. System Requirements

System Name: LibraryManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- The system works as follows: the user connects his PC to the projector and then starts the camera.
- The system detects the user's face using the YOLOv5 model.
- The system uses Background Subtraction (BS) to separate the moving parts for example here we mean the user's face.
- The system uses threading for concurrent execution of tasks within a single process.
- The system includes Gesture Recognition.
- The system includes Face Recognition.
- The system includes Hand Tracking.
- The system includes Single-Hand Interaction.
- The system includes customizing the actions for specific gestures.

Non-Functional Requirements

- Competitive price

Description: Our optimal goal is the price. After making a competitive analysis we figured out that our system is competitive.

- Real-time Collaboration

Description: Enable real-time collaboration by supporting interactions on the projected surface, fostering collaboration.

- Low-Latency

Description: The system must analyze gestures and reflect the corresponding action on the projected surface with low latency.

- Scalability

Description: The system should be able to handle multiple users and devices without compromising performance.

- Usability

Description: The system should be easy to use and navigate, with an intuitive interface that minimizes user error.

- Security

Description: The system should ensure the confidentiality, integrity, and availability of user data and

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, con

```
System LibraryManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component Camera = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Captures hand gesture commands and the user's facial features";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component YOLOv5Model = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Detects the user's face using the YOLOv5 model.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component BackgroundSubtraction(BS) = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Separates the moving parts (e.g., user's hand movement) from the background";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component HistogramEqualization = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Optimizes image contrast.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component Threading = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Enables concurrent execution of tasks within a single process";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component Multiprocessing = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Enables parallel execution of tasks across multiple processes";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component GestureRecognition = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Recognizes and interprets hand gestures.";
```

```

    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component FaceRecognition = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Recognizes and identifies the user's face.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component HandTracking = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Tracks the user's hand movements.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component Single-HandInteraction = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Allows the user to interact with the system using a single h
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component CustomGestureActions = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Customizes the actions for specific gestures.";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment Camera.out to YOLOv5Model.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment YOLOv5Model.out to BackgroundSubtraction(BS).in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment BackgroundSubtraction(BS).out to HistogramEqualization.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment HistogramEqualization.out to GestureRecognition.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment Camera.out to GestureRecognition.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment GestureRecognition.out to HandTracking.in {
    Property delivery : String = "at-least-once";

```

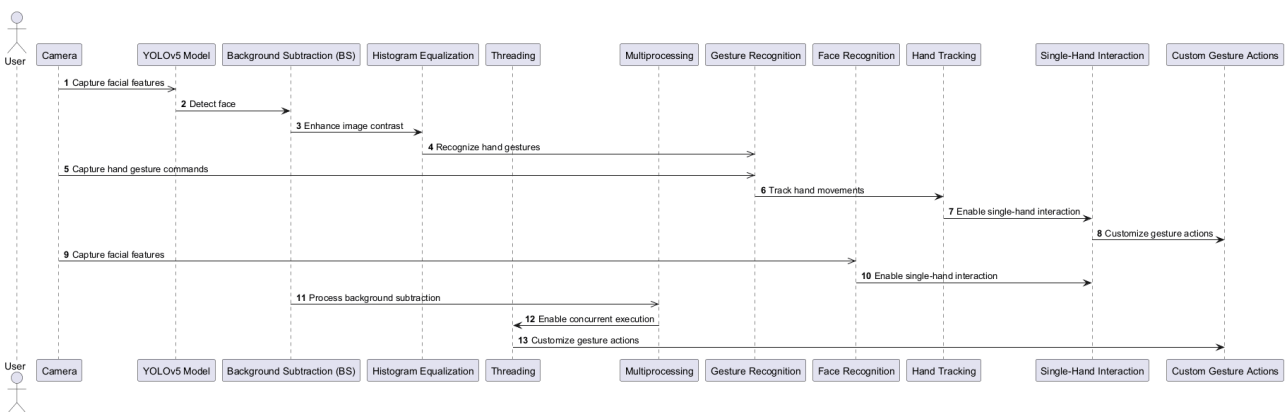
```

    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment HandTracking.out to Single-HandInteraction.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment Single-HandInteraction.out to CustomGestureActions.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment Camera.out to FaceRecognition.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment FaceRecognition.out to Single-HandInteraction.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment BackgroundSubtraction(BS).out to Multiprocessing.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment Multiprocessing.out to Threading.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment Threading.out to CustomGestureActions.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

3. Context View

5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

